

File Formats and Named Technology Associations — Detailed Lecture Notes

JKSSB Computer Awareness

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55.1 Extension and format are two different things

Every file stored on a computer follows some fixed plan for arranging its data. That plan is the **format** of the file. The format decides which program can open the file and what the file is able to store: a picture, a document, a table, or a database. An **image format** stores a picture; a **document format** stores a printable page; a **plain-text format** stores only characters.

A file name usually ends with an **extension** — the letters written after the last dot. In the name `report.png`, the extension is `png`. The extension is only a label; it tells the reader which format is expected inside. The extension and the format are therefore related but not the same thing. The exam tests this distinction directly, so it is stated here in its own section.

Definition (File format). *A fixed way of arranging the data in a file so that a particular program can read it, write it, and display it correctly.*

Definition (File extension). *The letters after the last dot in a file name, such as `png` in `report.png`. The extension signals the file's format.*

Term	Meaning in this lesson
Extension	The letters after the last dot in a file name.
Format	The actual arrangement of data inside the file.
PNG	Portable Network Graphics, an image format.
PDF	Portable Document Format, a fixed-layout document format.
CSV	Comma-Separated Values, a table stored as plain text.
TXT	A plain text file with no formatting.
.mdb / .accdb	Older and newer Microsoft Access database formats.

55.2 Image formats

The best-known image format for this course is **PNG**, which stands for **Portable Network Graphics**. PNG is a raster image format: it stores a picture as a grid of pixels. It uses *lossless* compression, meaning the picture is saved without permanently throwing away detail, and it supports transparency.

Other common image formats are **JPEG** (written `.jpg` or `.jpeg`), a *lossy* format often used for photographs, and **GIF** (`.gif`), which also supports simple animation. Typical image extensions are `.png`, `.jpg`, `.jpeg`, `.gif`, `.bmp`, and `.tiff`. The key exam point is simple: PNG, JPEG, and GIF are all **image formats**, and none of them is a document, spreadsheet, or database format.

Definition (Raster image). *An image stored as a grid of pixels. PNG and JPEG are raster formats.*

55.3 PDF: the Portable Document Format

PDF stands for **P**ortable **D**ocument **F**ormat, a format created by Adobe. Its defining property is a **fixed layout**: a PDF keeps the exact page layout, fonts, and formatting on every device, so the document looks the same whether it is opened on a desktop, a phone, or a printer. This is why PDF is the standard format for sharing a finished document that must not shift or reflow.

A PDF opens in a **PDF reader**, the best-known being **Adobe Acrobat Reader**. A **web browser** can also display a PDF, but reading a PDF is not the same as editing an editable word-processing document. A PDF may contain real, selectable text, or it may be a **scanned image** of each page. A scanned PDF is only a picture; to make its text searchable and selectable, **OCR** (Optical Character Recognition) must be applied. **PDF/A** is an archival PDF standard meant for long-term preservation.

Example 55.3.1. *A notice is scanned on a photocopier and saved as a PDF. The PDF now contains pictures of the pages, not typed text. Selecting a sentence with the mouse does nothing, and searching for a word finds nothing. Running OCR on the file creates a text layer, after which the same words can be searched and copied. The format is still PDF; only the internal content has changed from an image to searchable text.*

55.4 The Office document family

Microsoft Office files use a matching family of extensions that are easy to remember as a group.

Extension	Format	Program
.docx	Word document	Microsoft Word
.xlsx	Excel workbook	Microsoft Excel
.pptx	PowerPoint presentation	Microsoft PowerPoint

These are the modern **Office Open XML** formats. The older, pre-2007 extensions were **.doc** (Word), **.xls** (Excel), and **.ppt** (PowerPoint). For an exam, the pairing to hold is: **.docx = Word**, **.xlsx = Excel**, **.pptx = PowerPoint**; each extension always belongs to the same program, in the modern and the older naming alike.

55.5 Plain-data formats: CSV and TXT

CSV stands for **C**omma-**S**eparated **V**alues. A CSV file stores a table as plain text, with the values in each row separated by commas. Because it is only text, a CSV file can be opened by a spreadsheet program such as Excel, by a database, or by a simple text editor. The commas are the field separators; the file itself carries no formatting such as bold text or colours.

TXT denotes a **plain text** file. It stores only characters, with no formatting and no table structure. Both CSV and TXT are simple, vendor-neutral formats: unlike a **.docx** or **.xlsx** file, they do not belong to one program. The distinction to keep is that **CSV holds a table written as text separated by commas**, while **TXT holds plain characters with no inherent structure**.

55.6 Microsoft Access databases: .mdb and .accdb

Microsoft Access is the database program of the Office family, and it uses two extensions that candidates often confuse. **.mdb** is the **older** Access database format, used by Access 2003 and earlier. **.accdb** is the **newer** Access database format, used from Access 2007 onward. Both files can hold the standard database objects: tables, queries, forms, reports, macros, and modules.

The important point is that **both** **.mdb** and **.accdb** are **Microsoft Access database** files. Neither is a Word document, an Excel workbook, or a PowerPoint presentation.

55.7 Named technology associations

This section is the high-yield part of the lesson: matching a named technology to the organisation that produced it or the purpose it serves.

Firefox is a **web browser** developed by **Mozilla**. Mozilla is the organisation, and Firefox is its browser product; Mozilla also produces **Thunderbird**, an email client. Firefox must not be confused with Microsoft's browser, **Edge** (or the older Internet Explorer), nor with a PDF reader.

C-DAC stands for the **Centre for Development of Advanced Computing**. It is an Indian research and development organisation under the **Ministry of Electronics and Information Technology (MeitY)**. **PARAM** is a family of **supercomputers** developed by C-DAC; **PARAM 8000** was India's first indigenous supercomputer. The pairing to remember is **PARAM = Indian supercomputer, C-DAC = the organisation that developed it**.

55.8 Software roles: browser, PDF reader, and remote support

Software is often tested by the role it performs rather than by its maker.

Role	Example
Web browser	Firefox, Google Chrome, Microsoft Edge
PDF reader	Adobe Acrobat Reader
Remote-support tool	TeamViewer, AnyDesk, Chrome Remote Desktop

A **web browser** is used to visit web pages; a **PDF reader** is used to open PDF documents; a **remote-support tool** lets one computer control or view another over a network. Firefox is a browser and Adobe Acrobat Reader is a PDF reader; neither is a remote-support tool. Match the software to its *job*, not to a similar-sounding name.

Example 55.8.1. *A help-desk officer needs to see a distant user's screen to fix a problem. The right tool is a remote-support program such as TeamViewer or AnyDesk, not a browser and not a PDF reader. If the user then sends a manual as a fixed-layout file, it will arrive as a PDF and open in a PDF reader. If the user sends a spreadsheet of data, it will be an XLSX (Excel) or a CSV. Each task has its own format and its own tool.*

55.9 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. An **extension** is the letters after the dot; a **format** is how the data is stored. "Extension = format" is wrong (TRM-09, TRAP-34).

2. **PNG, JPEG, and GIF are image formats.** CSV is **not** an image format; it is a plain-text table format.
3. **PDF** is a fixed-layout document format. A scanned PDF is a picture and needs **OCR** to become searchable.
4. **.docx = Word, .xlsx = Excel, .pptx = PowerPoint;** the extension always matches the program.
5. **.mdb** and **.accdb** are **both Access database** files — older and newer respectively.
6. **Firefox is a browser by Mozilla; PARAM is a supercomputer by C-DAC.** Do not swap the makers.

Exercises

1. Distinguish a file *extension* from a file *format*, using **report.png** as the example.
2. Name any three image formats and state one property of PNG.
3. Explain what is meant by a fixed-layout format, and name the format created by Adobe that behaves this way.
4. Match each extension to its program: **.docx**, **.xlsx**, **.pptx**, **.accdb**.
5. What does CSV stand for, and how is its data arranged?
6. (a) Who develops Firefox? (b) What is PARAM, and which organisation develops it?

Solutions

1. The extension is the suffix after the last dot — here **png**. The format is the way the data inside the file is actually stored — here PNG, the Portable Network Graphics image format. The extension signals the format but is not the format itself.
2. PNG (Portable Network Graphics), JPEG (**.jpg/.jpeg**), and GIF (**.gif**), among others. PNG uses lossless compression and supports transparency.
3. A fixed-layout format keeps the exact page layout, fonts, and formatting on every device, so the document looks the same wherever it is opened. PDF (Portable Document Format) is the format created by Adobe that behaves this way.
4. **.docx** → Microsoft Word; **.xlsx** → Microsoft Excel; **.pptx** → Microsoft PowerPoint; **.accdb** → Microsoft Access (newer database format).
5. CSV stands for Comma-Separated Values. Its data is stored as a table in plain text, with the values in each row separated by commas.
6. (a) Firefox is developed by Mozilla. (b) PARAM is a family of supercomputers, developed by C-DAC (the Centre for Development of Advanced Computing); PARAM 8000 was India's first indigenous supercomputer.